

Financial Markets and Derivatives

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1 Introduction

1. **Financial Instrument:** A legal contract having monetary value. It's usually an asset on one entity and liability to other. Example: Check, Gold future contract, shares, bonds, personal loans, etc. Every financial security and product is a financial instrument but not the other way around.
2. **Financial Security:** A specific kind of financial instrument that is fungible and negotiable. Fungible means it can be swapped for another of the same kind, and negotiable means that it can be readily traded on private or public markets. Examples: Bonds, and Stocks.
3. **Financial Product:** Package or service created by a financial institution to help individuals save, invest, or protect their money. Examples: Life insurance policy, mutual funds, etc.
4. **Stock:** General term used to describe the ownership certificate of any firm.
5. **Share:** Smallest unit of a company's stock. Stock of the substance, and share is the unit.
6. **Systemic Risk:** Risk that a default by one financial institution will create a "ripple effect" that leads to defaults by other financial institutions and threaten the stability of the financial system.
7. **Forward Contract:** An agreement to buy or sell an asset at a certain future time for a certain price. Payoff from a long position in a forward contract is $S_T - K$.
8. **Futures Contracts:** Forward contract, but is standardized and traded on an exchange.
9. **Options:** There are two types of options:
 - (a) **Call Option:** Gives the holder the right to buy the underlying asset by a certain date for a certain price.
 - (b) **Put Option:** Gives the holder the right to sell the underlying asset by a certain date for a certain price.
10. American options can be exercised at any time up to the expiration date.
11. American and European here do not refer to the location of the option or the exchange.
12. Buyers are referred to as having *long positions*; sellers are referred to as having *short positions*.
13. Selling an option is also known as *writing the option*.
14. **Payoffs:**
 - (a) **Long CE:** $(S_T - K)^+$
 - (b) **Short CE:** $-(S_T - K)^+$
 - (c) **Long PE:** $(K - S_T)^+$
 - (d) **Short PE:** $-(K - S_T)^+$
15. **Arbitrage:** Riskless profit.
16. **Trading Volume:** Number of contracts traded in a day.
17. **Open Interest:** Number of contracts outstanding.
18. **Patterns of Future:**
 - (a) **Normal Market:** Futures market where futures prices increase with maturity.
 - (b) **Inverted Market:** Futures market where futures prices decrease with maturity.
19. **Orders:**
 - (a) **Market Order:** Request that a trade be carried out immediately at the best price available in the market.
 - (b) **Limit Order:** Specifies a particular price. Order can be executed only at this price or at one more favourable to the trader.
 - (c) **Stop Order/Stop-loss Order:** Also specifies a particular price. Order is executed at the best price once a bid or ask is made at that particular price or a less favourable price.
 - (d) **Stop-limit order:** Combination of a stop order and a limit order. Two prices must be specified: stop price, and limit price.

- (e) **Market-if-touched order/MIT:** Executed at the best available price after a trade occurs at a specified price or at a price more favourable than the specified price. An MIT becomes a market order once the specified price has been hit.

20. **CAPM:** Expected return on asset = $R_F + \beta(R_M - R_F)$

21. **Risk-Neutral World:** A world where investors are risk-neutral, i.e., they don't demand increased expected return to compensate for increased risk. This is in contrast with the real world, where people are risk-averse. Risk-neutral world has two features:

- (a) Expected rate of return on any investment is the risk-free rate.
- (b) The discounted rate used for an expected payoff on an instrument is the risk free rate.

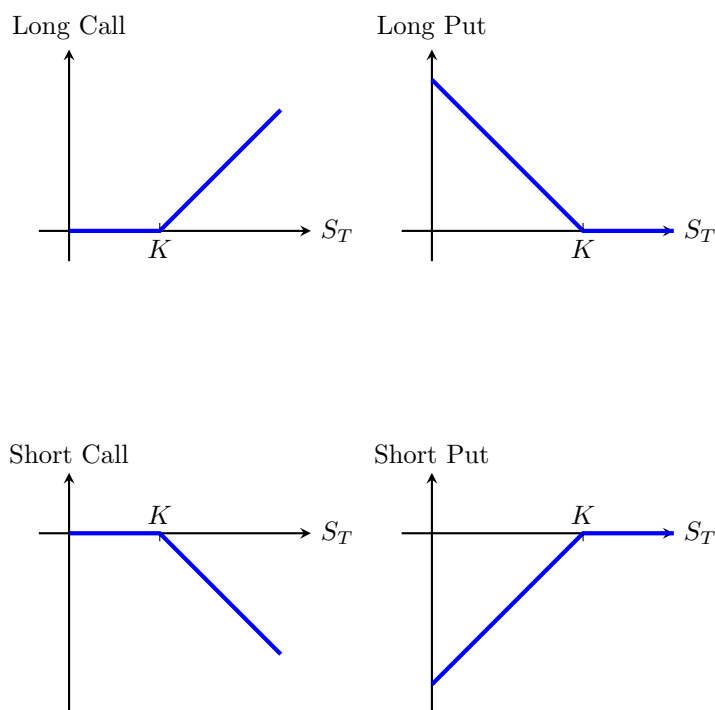


Figure 1: Payoff diagrams for the four basic option positions.

2 Interest Rates

1. **Interest rate:** The amount of money a borrower promises to pay the lender.
2. **Credit Risk:** Risk that there will be a default by the borrower. It strongly influences the IR charged for any transaction.
3. **Credit Spread:** Extra amount added to a risk-free interest rate to allow for the credit risk.
4. **Basis Points:** 1bps = 0.01%
5. **Types of Rates**
 - (a) **Treasury Rates:** Rates an investor earns on Treasury bills and Treasury Bonds.
 - (b) **Overnight Rates:** Weighted average of rates in brokered overnight transactions. Financial institutions engage in overnight transactions as they have to maintain a reserve in their central bank accounts.
 - (c) **Repo Rates:** Repurchase agreement rates. A financial institution agrees to sell its securities for a certain price and buy them back at a later time for a slightly higher price. The difference in price is the interest, called the repo rate.
 - i. **Overnight Repo:** Funds are lent overnight.
 - ii. **Term Repo:** Longer-term arrangement.
 - iii. **SOFR:** Secured overnight financing rate. Volume-weighted average of the rates on overnight repo transactions in the US.
6. **Reference Rates**
 - (a) **LIBOR:** London Interbank Offered Rate. Estimates of unsecured borrowing rates for creditworthy banks.
 - (b) **SOFR:** Reference rate in the US.
 - (c) **SONIA:** Stirling Overnight Index Average. Reference rate in UK.
 - (d) Longer rates such as three-month rates, six-month rates, or one-year rates can be determined from overnight rates by compounding them daily.
 - (e) If the annualized SOFR on the i th business day of a period is r_i ($1 \leq i \leq n$) and the rate applies to d_i days. The annualised interest rate for the period is
$$\left(\prod_{i=1}^n \left(1 + \frac{r_i \cdot d_i}{360} \right) - 1 \right) \frac{360}{D}$$
Where d_i is the number of days the rate applies. (1 for weekdays, and 3 for Fridays), and D is the total days in the period.
 - (f) LIBORs are forward looking as they are determined at the beginning of the period to which they apply. SOFR, SONIA, etc are backward looking.
7. **Compound Interest:** $A(1 + R)^n$. And if the rate is compounded m times per annum, the terminal value of the investment is $A(1 + R/m)^{mn}$.
8. **Continuous Compounding:** The limit as the compounding frequency m , tends to infinity. An amount A invested for n years at rate R grows to Ae^{Rn} .
9. **Zero Rates:** The n -year zero-coupon interest rate is the rate of interest earned on an investment that starts today and lasts for n years. All the interest and principal is realized at the end of n years. This is also called as the n -year spot rate, the n -year zero rate, or just the n -year zero.
10. **Bond Pricing:** Most bonds pay coupons to the holder periodically. The bond's principal (also known as its par or face value) is paid at the end of its life. The price can be calculated as the present value of all the cash flows that will be received by the owner of the bond. Sometimes bond traders use the same discount rate for all the cash flows, but a more accurate approach is to use a different zero rate for each cash flow.
11. **Bond Yield:** The single discount rate that, when applied to all cash flows, gives a bond price equal to its market price.
12. **Par Yield:** The coupon rate that causes the bond price to equal its par value.

13. **Determining Zero Rates:** Using the bootstrap method. We know the zero rates for shorter maturities and then iteratively use them to calculate the zero rates for longer maturities, using the PV equation.
14. **Zero Curve:** Zero rates as a function of maturities. We interpolate to find the rates for maturities for which bonds are not available.
15. **Forward Rates:** Forward interest rates are the rates of interest implied by current zero rates for periods of the time in the future. If R_1 and R_2 are the zero rates for maturities T_1 and T_2 , and R_F is the forward interest rate for the period of time between T_1 and T_2 , then

$$R_F = \frac{R_2 T_2 - R_1 T_1}{T_2 - T_1}$$

The above equation can also be written as:

$$R_F = R_2 + (R_2 - R_1) \frac{T_1}{T_2 - T_1}$$

Now, taking the limit as T_2 approaches T_1 , we get

$$R_F = R + T \frac{\partial R}{\partial T}$$

where R is the zero rate of a maturity of T .

16. The above quantity is called the **instaneous forward rate** of a maturity of T . This is the forward rate that is applicable to a very short future time period that begins at time T .
17. **Forward Rate Agreements:** An FRA is an agreement to exchange a predetermined fixed rate for a reference rate that will be observed in the market at a future time. Both rates are applied to a specified principal but the principal itself is not exchanged. When an FRA is first set up the fixed rate is normally equal to the forward rate, so that its value is zero. As time progresses, the value is liable to change.
18. **Duration:** Measure of how long the holder of the bond has to wait before receiving the present value of the cash payments. A zero coupon bond has a duration equal to its maturity. However, a coupon bearing bond has a duration less than its maturity.
19. Duration of a bond, D is defined as

$$D = \sum_{i=1}^n t_i \left[\frac{c_i e^{-y t_i}}{B} \right]$$

20. The duration is therefore a weighted average of the times when payments are made, with the weight applied to time t_i being equal to the proportion of the bond's total PV provided by the cashflow at time t_i .
21. Since the bond price, B is a function of the yield, y , we have,

$$\Delta B = \frac{dB}{dy} \Delta y$$

Also,

$$B = \sum_{i=1}^n c_i e^{-y t_i}$$

Thus,

$$\Delta B = -\Delta y \sum_{i=1}^n c_i t_i e^{-y t_i}$$

Note the negative relationship between B and y . Also, the above equation reduces to

$$\Delta B = -BD\Delta y \implies \frac{\Delta B}{B} = -D\Delta y$$

22. **Modified Duration:** The preceding analysis is based on the assumption that y is expressed with continuous compounding. If y is expressed with annual compounding, the relationship becomes,

$$\Delta B = -\frac{BD\Delta y}{1 + y}$$

More generally, if y is expressed with a compounding frequency of m times per year, then

$$\Delta B = -\frac{BD\Delta y}{1 + y/m}$$

The variable, D^* , defined by

$$D^* = \frac{D}{1 + y/m}$$

is referred to as the bond's *modified duration*. It allows the relation to be simplified to

$$\Delta B = -BD^*\Delta y$$

23. The duration of a bond portfolio can be defined as a weighted average of the durations of the individual bonds in the portfolio, with the weights being proportional to the bond prices.
24. **Convexity:** The duration relationship applies only to small changes in yields. For large changes in yields, the percentage change in value of B curve has a curvature. *Convexity* measures this curvature.

$$C = \frac{1}{B} \frac{d^2 y}{dy^2} = \frac{\sum_{i=1}^n c_i t_i^2 e^{-y t_i}}{B}$$

From Taylor series, we obtain a more accurate expression, given by

$$\Delta B = \frac{dB}{dy} \Delta y + \frac{1}{2} \frac{d^2 B}{dy^2} \Delta y^2$$

This leads to

$$\frac{\Delta B}{B} = -D\Delta y + \frac{1}{2}C(\Delta y)^2$$

25. For a portfolio with a particular duration, the convexity of a bond portfolio tends to be greatest when the portfolio provides payments evenly over a long period of time. It is least when the payments are concentrated around one particular point of time.

3 Swaps

1. **Swap:** An OTC agreement between two parties to exchange cashflows in future.
2. Forward contract is a simple example of swap.
3. **Interest Rate Swap:** A swap where interest at a predetermined fixed rate, applied to a certain principal, is exchanged for interest at a floating reference rate, applied to the same principal, with regular exchanges being made for an agreed period of time.
4. **Overnight Indexed Swap:** An OIS is an agreement to exchange a fixed rate of interest for a reference rate of interest that is calculated from realized overnight rates.
5. **Reasons for trading Interest Rate Swaps:**
 - (a) **Using the swap to transform a liability**
 - (b) **Using the swap to transform an asset**
6. **Comparative Advantage Argument:** Some companies have a comparative advantage when borrowing in fixed-rate markets, whereas other companies have a comparative advantage when borrowing in floating-rate markets. This difference can be exploited by two firms favoured in different markets by getting into an IRS.
7. **Valuation of IRS:** An IRS is worth close to zero when it is first initiated. But it is liable to change as time passes. It can be valued by considering the present value of the future cashflows using forward rates.
8. **Fixed-for-Fixed Currency Swaps:** Exchanging principal and interest payments at a fixed rate in one currency for principal and interest payments at a fixed rate in another country.
9. **Other Currency Swaps:**
 - (a) **Fixed-for-floating**
 - (b) **Floating-for-floating**
10. **Credit Default Swaps:** A CDS is like an insurance contract that pays off if a particular company or country defaults. The company or the country is known as *reference entity*.

4 Properties of Stock Options

1. Factors affecting Option Prices

- (a) The current stock price, S_0
- (b) The strike price, K
- (c) The time to expiration, T
- (d) The volatility of the stock price, σ
- (e) The risk-free interest rate, r
- (f) The dividends that are expected to be paid.

2. The direction of changes are summarised in the table below:

Table 4.1 Summary of the effect on the price of a stock option of increasing one variable while keeping all others fixed.

<i>Variable</i>	<i>European call</i>	<i>European put</i>	<i>American call</i>	<i>American put</i>
Current stock price	+	−	+	−
Strike price	−	+	−	+
Time to expiration	?	?	+	+
Volatility	+	+	+	+
Risk-free rate	+	−	+	−
Amount of future dividends	−	+	−	+

+ indicates that an increase in the variable causes the option price to increase or stay the same;

− indicates that an increase in the variable causes the option price to decrease or stay the same;

? indicates that the relationship is uncertain.

3. Assumptions for this chapter

- (a) There are no transaction costs.
- (b) All trading profits are subject to the same tax rate.
- (c) Borrowing and lending are possible at the risk-free interest rate.

4. Notations

- (a) S_0 : Current stock price
- (b) K : Strike price of option
- (c) T : Time to expiration of option
- (d) S_T : Stock price on the expiration date
- (e) r : Continuously compounded risk-free rate of interest for an investment maturing in time T
- (f) C : Value of an American call option
- (g) P : Value of an American put option
- (h) c : Value of an European call option
- (i) p : Value of an European put option

5. Bounds

- (a) $\max(S_0 - Ke^{-rT}, 0) \leq c \leq C \leq S_0$
- (b) $\max(Ke^{-rT} - S_0, 0) \leq p \leq Ke^{-rT}$
- (c) $\max(K - S_0, 0) \leq P \leq K$
- (d) $c \geq \max(S_0 - D - Ke^{-rT}, 0)$
- (e) $p \geq \max(D + Ke^{-rT} - S_0, 0)$

6. Put-Call Parity:

- (a) $c + Ke^{-rT} = p + S_0$
- (b) $c + D + Ke^{-rT} = p + S_0$

5 Trading Strategies Involving Options

Bull Spread (long call K_1 , short call K_2 , $K_1 < K_2$)

$$\text{Payoff} = \begin{cases} 0 & S_T < K_1 \\ S_T - K_1 & K_1 \leq S_T < K_2 \\ K_2 - K_1 & S_T \geq K_2 \end{cases}$$

Bear Spread (long put K_2 , short put K_1 , $K_1 < K_2$)

$$\text{Payoff} = \begin{cases} K_2 - K_1 & S_T \leq K_1 \\ K_2 - S_T & K_1 < S_T < K_2 \\ 0 & S_T \geq K_2 \end{cases}$$

Box Spread (bull call spread + bear put spread, $K_1 < K_2$)

$$\text{Payoff} = K_2 - K_1 \quad \forall S_T$$

Butterfly Spread (long K_1 , short $2 \times K_2$, long K_3 ; $K_1 < K_2 < K_3$)

$$\text{Payoff} = \begin{cases} 0 & S_T \leq K_1 \\ S_T - K_1 & K_1 < S_T \leq K_2 \\ K_3 - S_T & K_2 < S_T < K_3 \\ 0 & S_T \geq K_3 \end{cases}$$

Straddle (long call + long put, same strike K)

$$\text{Payoff} = \begin{cases} K - S_T & S_T < K \\ 0 & S_T = K \\ S_T - K & S_T > K \end{cases} = |S_T - K|$$

Strangle (long put K_1 , long call K_2 , $K_1 < K_2$)

$$\text{Payoff} = \begin{cases} K_1 - S_T & S_T < K_1 \\ 0 & K_1 \leq S_T \leq K_2 \\ S_T - K_2 & S_T > K_2 \end{cases}$$

See book for more diagrams, etc.

6 The Black-Scholes-Merton Model

Black-Scholes-Merton Differential Equation

$$\frac{\partial f}{\partial t} + rS \frac{\partial f}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 f}{\partial S^2} = rf$$

Pricing Formulas

$$\begin{aligned}c &= S_0 N(d_1) - K e^{-rT} N(d_2) \\p &= K e^{-rT} N(-d_2) - S_0 N(-d_1) \\d_1 &= \frac{1}{\sigma \sqrt{T}} (\ln(S_0/K) + (r + \sigma^2/2)T) \\d_2 &= \frac{1}{\sigma \sqrt{T}} (\ln(S_0/K) + (r - \sigma^2/2)T) = d_1 - \sigma \sqrt{T}\end{aligned}$$

7 The Greeks

$$\begin{aligned}\Delta_{\text{call}} &= N(d_1) \\\Delta_{\text{put}} &= N(d_1) - 1 \\\Gamma &= \frac{N'(d_1)}{S_0 \sigma \sqrt{T}} \\\nu &= S_0 \sqrt{T} N'(d_1) \\\rho_{\text{call}} &= K T e^{-rT} N(d_2) \\\rho_{\text{put}} &= -K T e^{-rT} N(-d_2) \\\Theta_{\text{call}} &= -\frac{S_0 N'(d_1) \sigma}{2 \sqrt{T}} - r K e^{-rT} N(d_2) \\\Theta_{\text{put}} &= -\frac{S_0 N'(d_1) \sigma}{2 \sqrt{T}} + r K e^{-rT} N(-d_2) \\N'(x) &= \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \\\Delta \Pi &\approx \Theta \Delta t + \frac{1}{2} \Gamma (\Delta S)^2 \\\Theta + r S \Delta + \frac{1}{2} \sigma^2 S^2 \Gamma &= r \Pi\end{aligned}$$